

Shared PPO Update Core

The one reusable clipped-PPO optimizer; all continual weighting enters via actor coefficients.

Overview

One update is `ppo_epochs` passes of minibatch SGD over one or more collected rollout STREAMS (one per task, each already GAE-processed). Each stream i adds $c_i \cdot (\mathrm{clipped_surrogate}_i)$ to the actor loss, a STANDARD $\mathrm{vf_coef} \cdot (\mathrm{value_MSE}_i)$ critic loss, and an entropy bonus $-\mathrm{ent_coef} \cdot \mathcal{H}_i$; the sum is optimized under a single global grad-norm clip. Only the list of actor coefficients c_i differs between the local phase ($[1]$) and the constrained global phase ($[\omega_1, \dots, \omega_{k-1}, 2\mu s_k]$).

KEY STABILIZER: the actor coefficients are renormalized by their sum before use. The dual term $2\mu s_k$ is unbounded in μ ; without normalization a large μ makes the actor gradient dominate the shared grad-norm clip and STARVES the shared critic's value gradient, breaking GAE. Division by the sum is a positive rescaling of the ascent direction: it preserves the relative past/current weighting (and the saddle fixed point) and only bounds the step magnitude. For the local phase ($c = [1]$) it is a no-op.

Algorithm 1: `optimize_batches(policy, opt, streams, actor_coeffs)`

Input: shared-trunk multi-head actor+critic policy; Adam optimizer `opt` over its trainable params

Input: streams: list of GAE rollout batches (obs, actions, old log-probs, advantages, returns, task-id), all

Input: `actor_coeffs` = $[c_1, \dots, c_m]$: one actor weight per stream

Input: **config:** `num_minibatches`, `ppo_epochs`, `clip_ratio`, `vf_coef`, `ent_coef`, `max_grad_norm`; optional (π_L , `bc_coef`)

Input: optional KL early-stop flag `kl_stop` (used by the local phase; the global phase runs full epochs)

Output: in-place gradient steps on policy

Output: diagnostics: mean pg-loss, value-loss, entropy, approx-KL, clip-fraction

1 $N \leftarrow \text{rows in streams}[0]$; $\text{mb} \leftarrow \max(1, N/\text{num_minibatches})$

2 $S \leftarrow \sum_i c_i$; if $S > 0$: renormalize $c_i \leftarrow c_i/S$ for all i

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3 for epoch = 1, ..., ppo_epochs do
4     draw a fresh random permutation of  $\{1, \dots, N\}$  (shared by all streams)
5     for each minibatch index set idx (size mb) do
6          $L \leftarrow 0$ 

7         for each  $(c_i, \text{batch}_i)$  in zip(actor_coeffs, streams) do
8             dist,  $V \leftarrow$  policy.dist_value(batchi.obs[idx], batchi.task_id)

9              $\rho \leftarrow \exp(\text{dist.log} p(a) - \text{old log } p)$ ;  $A \leftarrow$  advantages (optionally standardized)

10             $\text{pg}_i \leftarrow \text{mean max}(-A\rho, -A \cdot \text{clip}(\rho, 1 - \text{clip\_ratio}, 1 + \text{clip\_ratio}))$ 

11             $\text{vloss}_i \leftarrow \frac{1}{2} \text{mean}(V - \text{returns})^2$ ;  $\mathcal{H}_i \leftarrow \text{mean dist.entropy}()$ 

12             $L \leftarrow L + c_i \text{pg}_i + \text{vf\_coef} \cdot \text{vloss}_i - \text{ent\_coef} \cdot \mathcal{H}_i$ 

13            if bc set and this is the LAST (current-task) stream:  $L \leftarrow L + \text{bc\_coef} \cdot \text{mean KL}(\pi_L \| \text{dist})$ 

14            accumulate diagnostics (pg, vloss, entropy, approx-KL =  $\text{mean}(\rho - 1 - \log \rho)$ , clip-frac)
            end for
15            opt.zero-grad;  $L$ .backward
16            clip-grad-norm(policy.parameters(), max_grad_norm) (single GLOBAL clip)
17            opt.step
            end for
18        if kl_stop and target_kl > 0 and epoch-mean approx-KL > target_kl: break
        end for
19 return averaged {pg-loss, value-loss, entropy, approx-KL, clip-frac}

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What is updated, and when

The policy parameters move once per minibatch

(ppo_epochs $\times$ num_minibatches steps per update). Only params with

requires-grad move (the whole net normally; only the heads under the head-only probe). The

critic and entropy terms are IDENTICAL for every stream (coefficients `vf_coef` and `ent_coef`); ONLY the actor term carries the per-stream continual weight c_i . The KL early-stop is used by the local phase (`kl_stop = true`); the global consolidation phase passes `kl_stop = false` so the retention update is never throttled.